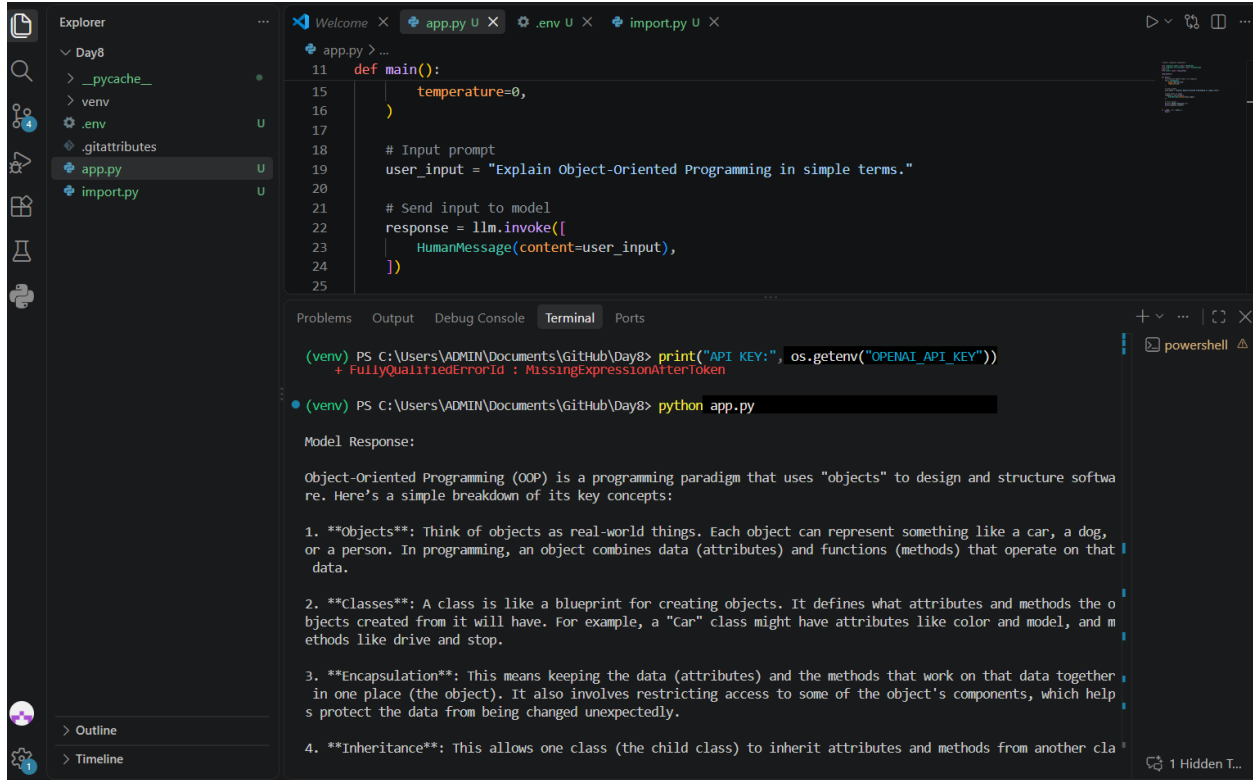


Day3 : Assignment - Create an app using Langchain and Tracing via Langsmith

Code: moved into Github Repository.

Result screenshot:



The screenshot shows a VS Code editor with a file explorer on the left containing files like `__pycache__`, `venv`, `.env`, `.gitattributes`, `app.py`, and `import.py`. The main editor window displays the `app.py` file with the following code:

```
11 def main():
12     temperature=0,
13 )
14
15 # Input prompt
16 user_input = "Explain Object-Oriented Programming in simple terms."
17
18 # Send input to model
19 response = llm.invoke([
20     HumanMessage(content=user_input),
21 ])
22
23 
```

Below the code, the terminal window shows the execution output:

```
(venv) PS C:\Users\ADMIN\Documents\Github\Day8> print("API KEY:", os.getenv("OPENAI_API_KEY"))
+ FullyQualifiedErrorId : MissingExpressionAfterToken

(venv) PS C:\Users\ADMIN\Documents\Github\Day8> python app.py

Model Response:

Object-Oriented Programming (OOP) is a programming paradigm that uses "objects" to design and structure software. Here's a simple breakdown of its key concepts:

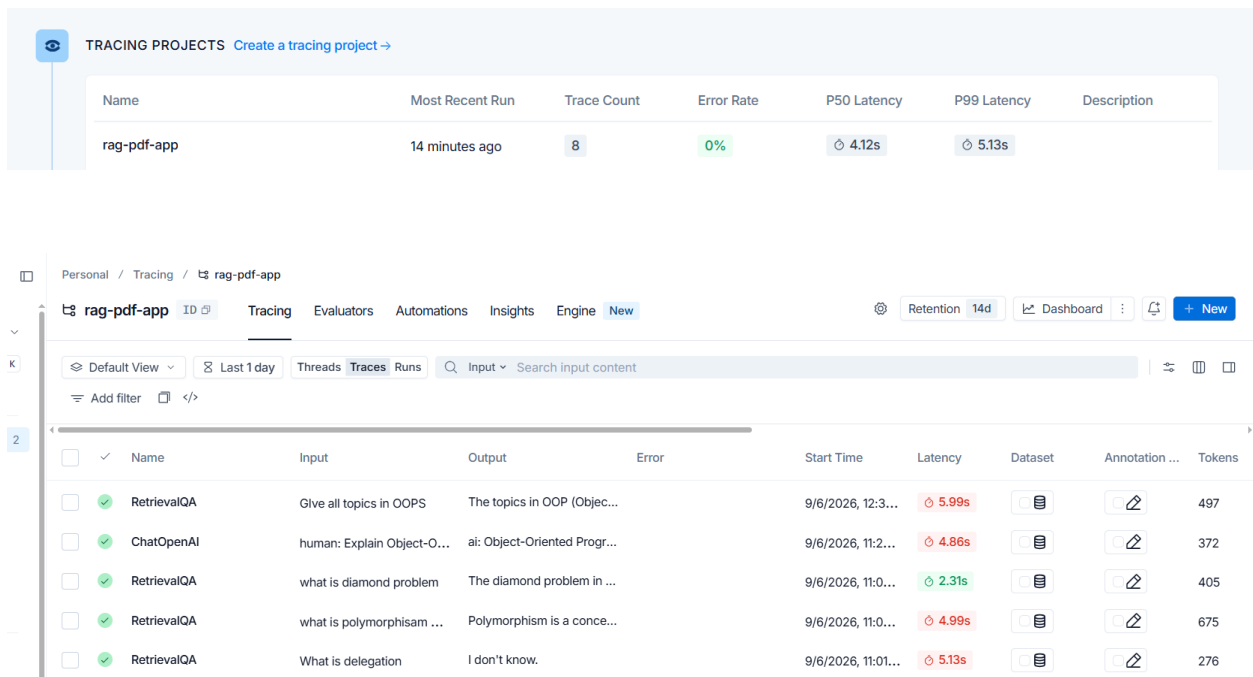
1. **Objects**: Think of objects as real-world things. Each object can represent something like a car, a dog, or a person. In programming, an object combines data (attributes) and functions (methods) that operate on that data.

2. **Classes**: A class is like a blueprint for creating objects. It defines what attributes and methods the objects created from it will have. For example, a "Car" class might have attributes like color and model, and methods like drive and stop.

3. **Encapsulation**: This means keeping the data (attributes) and the methods that work on that data together in one place (the object). It also involves restricting access to some of the object's components, which helps protect the data from being changed unexpectedly.

4. **Inheritance**: This allows one class (the child class) to inherit attributes and methods from another class (the parent class).
```

Tracing Screenshot:



The screenshot displays the LangSmith Tracing interface for the 'rag-pdf-app' project. The top section shows a table of tracing runs with columns: Name, Most Recent Run, Trace Count, Error Rate, P50 Latency, P99 Latency, and Description. Below this, the 'Tracing' tab is selected, showing a list of runs with columns: Name, Input, Output, Error, Start Time, Latency, Dataset, Annotation, and Tokens.

Name	Most Recent Run	Trace Count	Error Rate	P50 Latency	P99 Latency	Description
rag-pdf-app	14 minutes ago	8	0%	4.12s	5.13s	

Name	Input	Output	Error	Start Time	Latency	Dataset	Annotation	Tokens
RetrievalQA	Give all topics in OOPS	The topics in OOP (Objec...		9/6/2026, 12:3...	5.99s			497
ChatOpenAI	human: Explain Object-O...	ai: Object-Oriented Progr...		9/6/2026, 11:2...	4.86s			372
RetrievalQA	what is diamond problem	The diamond problem in ...		9/6/2026, 11:0...	2.31s			405
RetrievalQA	what is polymorphisam ...	Polymorphism is a conce...		9/6/2026, 11:0...	4.99s			675
RetrievalQA	What is delegation	I don't know.		9/6/2026, 11:01...	5.13s			276

LangSmith

Personal / Tracing / rag-pdf-app

rag-pdf-app ID

Tracing

Default View Last 1 day

VectorStoreRetriever Retriever

Time

Start 09/06/2026, 12:30:51 PM GMT+5:30

End 09/06/2026, 12:30:52 PM GMT+5:30

Tags

OpenAIEmbeddings FAISS

RetrievalQA

RetrievalQA

RetrievalQA 12:30 PM

5.99s 497 / \$0.0002

VectorStoreRetriever 1.50s

ChatOpenAI gpt-4o-mini

4.36s 497

Some runs have been hidden. [Show 2 hidden runs](#)

Feedback Input Output Attributes

Feedback

+ Add feedback

Input

Fields

query Give all topics in OOPS

Output

Fields

LangSmith

Personal / Tracing / rag-pdf-app

rag-pdf-app ID

Tracing

Default View Last 1 day

Add filter </>

ChatOpenAI LLM • gpt-4o-mini

Time

Start 09/06/2026, 12:30:52 PM GMT+5:30

End 09/06/2026, 12:30:57 PM GMT+5:30

Total cost breakdown

Input	30%	312	<\$0.0001
Output	70%	185	\$0.0001
Total	100%	497	\$0.0002

RetrievalQA

RetrievalQA 12:30 PM

5.99s 497 / \$0.0002

VectorStoreRetriever 1.50s

ChatOpenAI gpt-4o-mini

4.36s 497

Some runs have been hidden. [Show 2 hidden runs](#)

Feedback Input Output Attributes

Feedback

+ Add feedback

Input

Fields

query Give all topics in OOPS

Output

Fields

Messages Beta Details

Notes:

DAY 11

WEEK 2

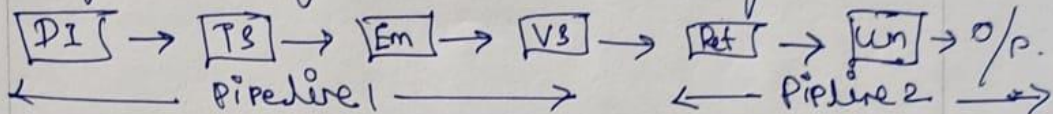
Topic: Introduction to RAG.

Takeaways

RAG → Potential Augmented Generation.
↳ AI models first searches for relevant information
RAG → Pipeline 1 and 2
Improve the basic RAG in a different way to
give better and more accurate.

Summary

They covered rag flow and types of rag improvement



Types of Rag → Self Rag, Corrective Rag, Fusion Rag
Bm 25 + re-rank

Today's Task

SUBMISSION STATUS	NOTES / REMARKS
☒ Submitted Task completed and shared	Understand Rag and the Pipeline Steps, as well as learned about different type of Rag for better accurate results.
☐ Not Submitted Will catch up tomorrow	

Tick the box only when it's true. Lying to yourself is the only real failure.

Code and Assignment Details: <https://github.com/JeyaprakashCyan/Day8>